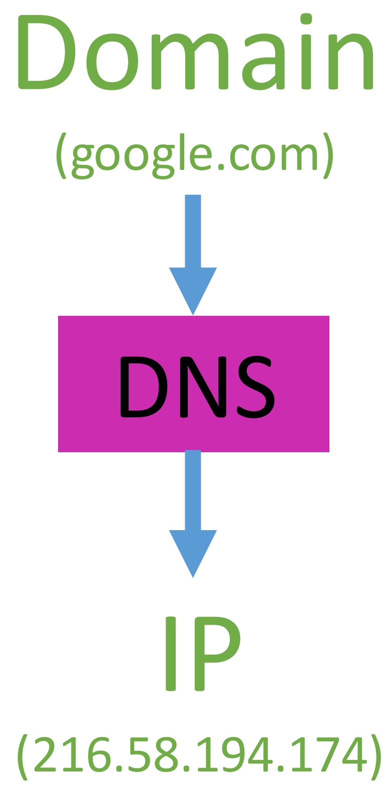
Domain Name System Forensics

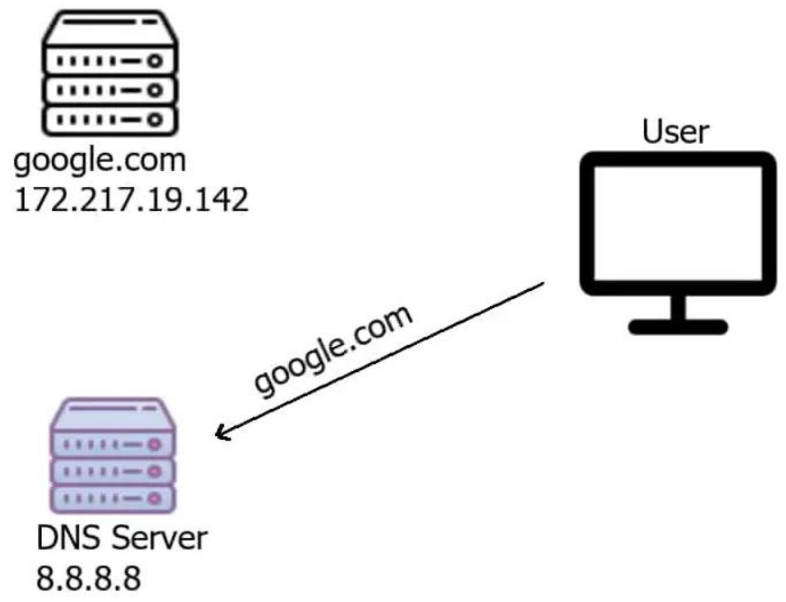
Domain Name System (DNS)

- Motivation: Every device is identified by an IP address, which is a unique numerical identifier.
- Problem: However, IP addresses are numbers and can be difficult for people to remember and use.
- Solution
  - Each device has a human-readable string (domain name). Each domain name corresponds to an IP address
  - DNS translates domain names into IP addresses and vice versa.

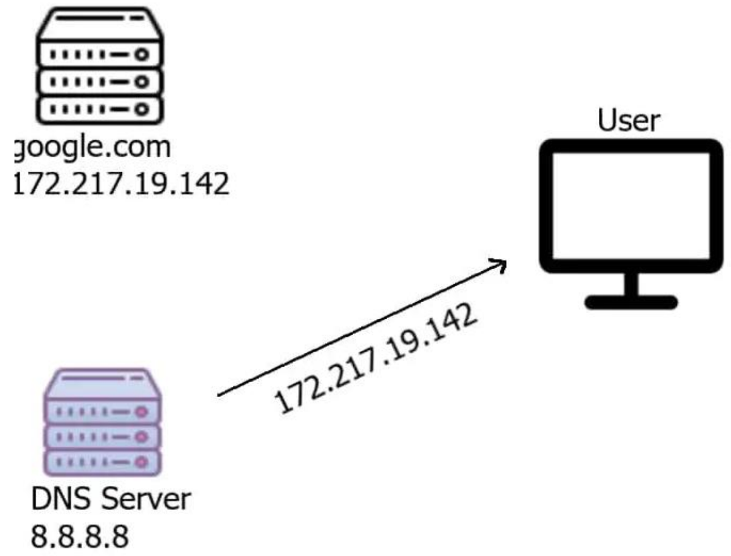


DNS like a phone book for the internet

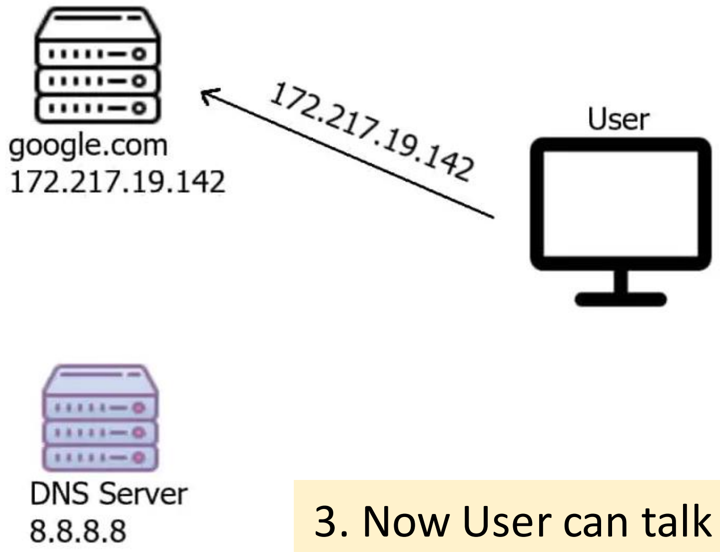
- Just like how you can look up your friend's phone number in a phone book to call them
  - your computer or tablet uses DNS to look up the address of a website you want to visit on the internet.
- When you type in the name of a website like YouTube
  - your computer or tablet sends a message to a special computer called a DNS server.
  - The DNS server looks up the address of YouTube and sends it back to your computer or tablet.
  - Then, your computer or tablet uses that address to find YouTube and show you all your favorite videos!
- So, the next time you use the internet, remember that DNS is working behind the scenes to make it all possible!



1. What is the IP of google.com?



2. Here is the IP of google.com

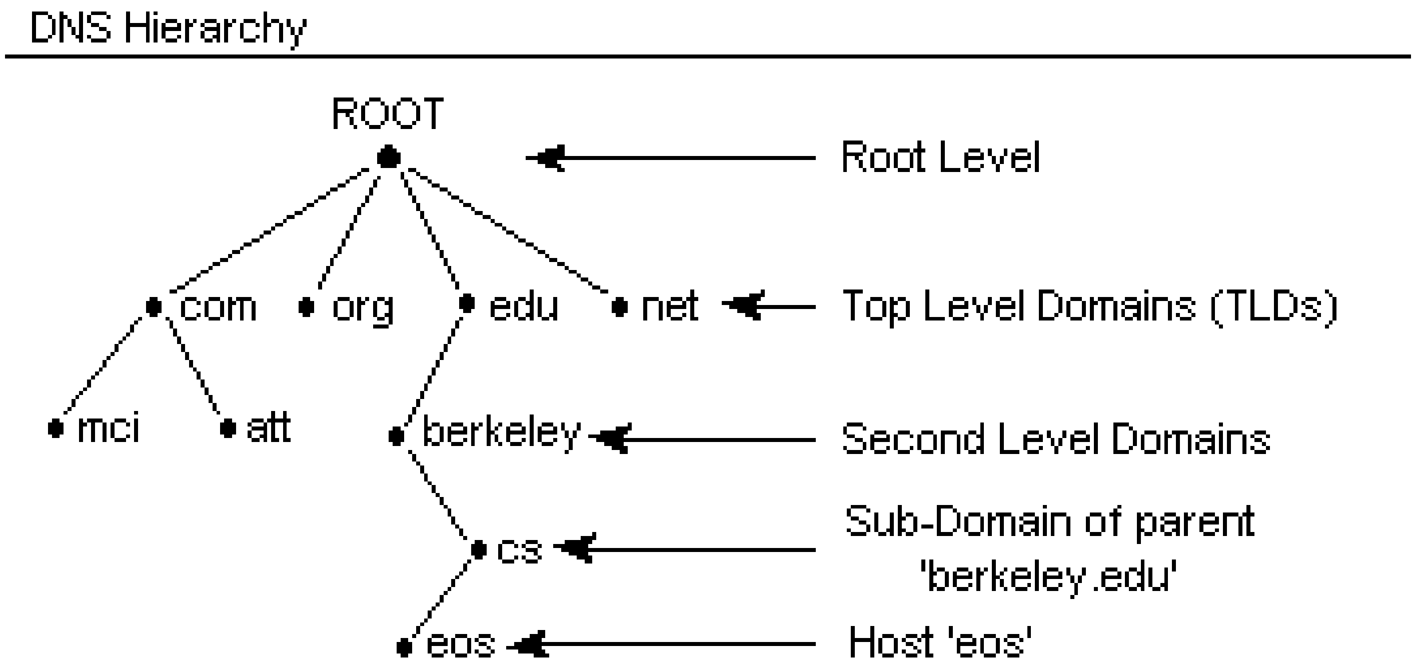


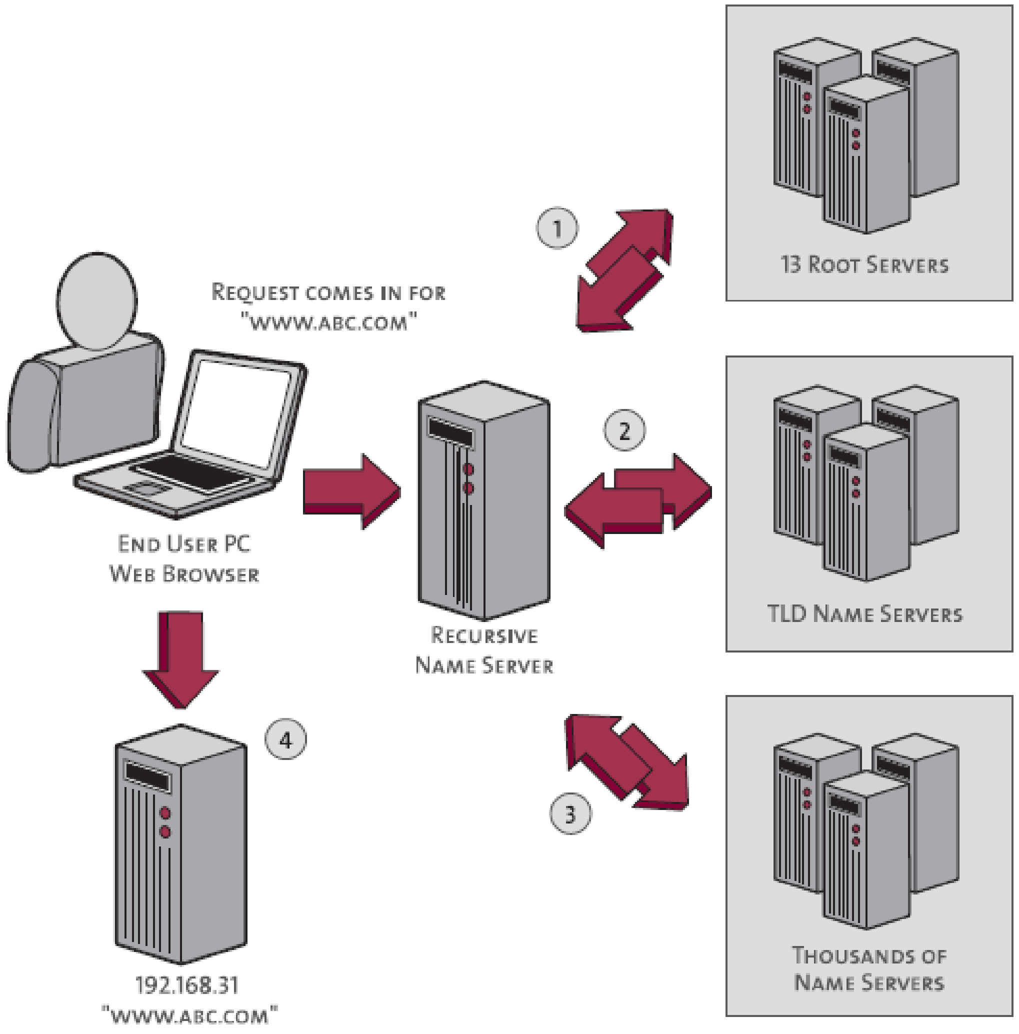
3. Now User can talk to google.com

	OSI Layers	Example Protocols
Media Layers	Application Layer - *Data*	HTTP, FTP, IRC, SSH, $\boxed{\text{DNS}}$
	Presenation Layer - *Data*	SSL, FTP, IMAP, SSH
	Session Layer - *Data*	VARIOUS, API'S, SOCKETS
	Transport Layer - *Segments*	TCP, UDP, ECN, SCTP, DCCP
Media Layers	Network Layer - *Packets*	IP, IPSec,ICMP, IGMP
	Data Link Layer - *Frames*	Ethernet, SLLIP, PPP, FDDI
	Physical Layer - *Bits*	Coax, Fiber, Wireless

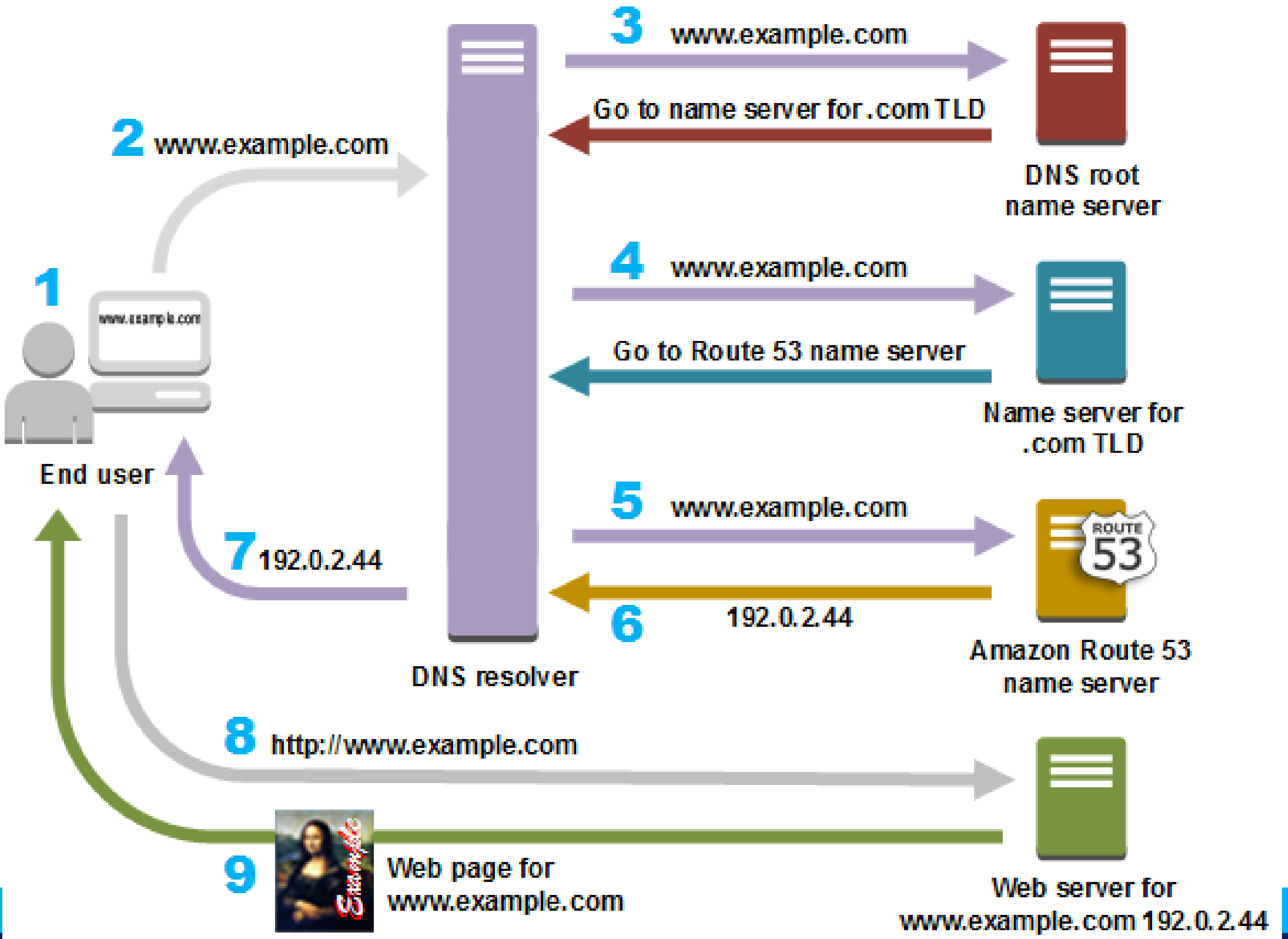
https://asmed.com/comptia-network-osi-model/

DNS is a hierarchical naming system





DNS Example



- DNS resolver is often managed by the user's Internet service provide
- TLD: top-level domain, such .com, org

https://aws.amazon.com/route53/what-is-dns/

Observe DNS with `dig`

dig

- It is a network administration tool used in Linux and other Unix-based operating systems to perform DNS (Domain Name System) queries.
  - retrieve information about DNS records, including
    - IP addresses associated with domain names,
    - mail server information, and
    - other DNS-related information.
- It sends a query to a specified DNS server and returns the response in a human-readable format.
  - It can be used to troubleshoot DNS issues, test DNS configuration changes, and gather information about DNS servers.

Create a working directory

```
┌──(student㉿kalit01)-[~]
└─$ mkdir dns

┌──(student㉿kalit01)-[~]
└─$ cd dns
```

student@kali01: ~/dns 105x27

```
┌──(student㉿kalit01)-[~/dns]
└─$ dig google.com

; <<>> DiG 9.18.12-1-Debian <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 11464
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             25      IN      A       142.251.163.113
google.com.             25      IN      A       142.251.163.139
google.com.             25      IN      A       142.251.163.102
google.com.             25      IN      A       142.251.163.100
google.com.             25      IN      A       142.251.163.138
google.com.             25      IN      A       142.251.163.101

;; Query time: 4 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Mar 13 10:01:02 EDT 2023
;; MSG SIZE  rcvd: 135
```

2. Generating dns traffic

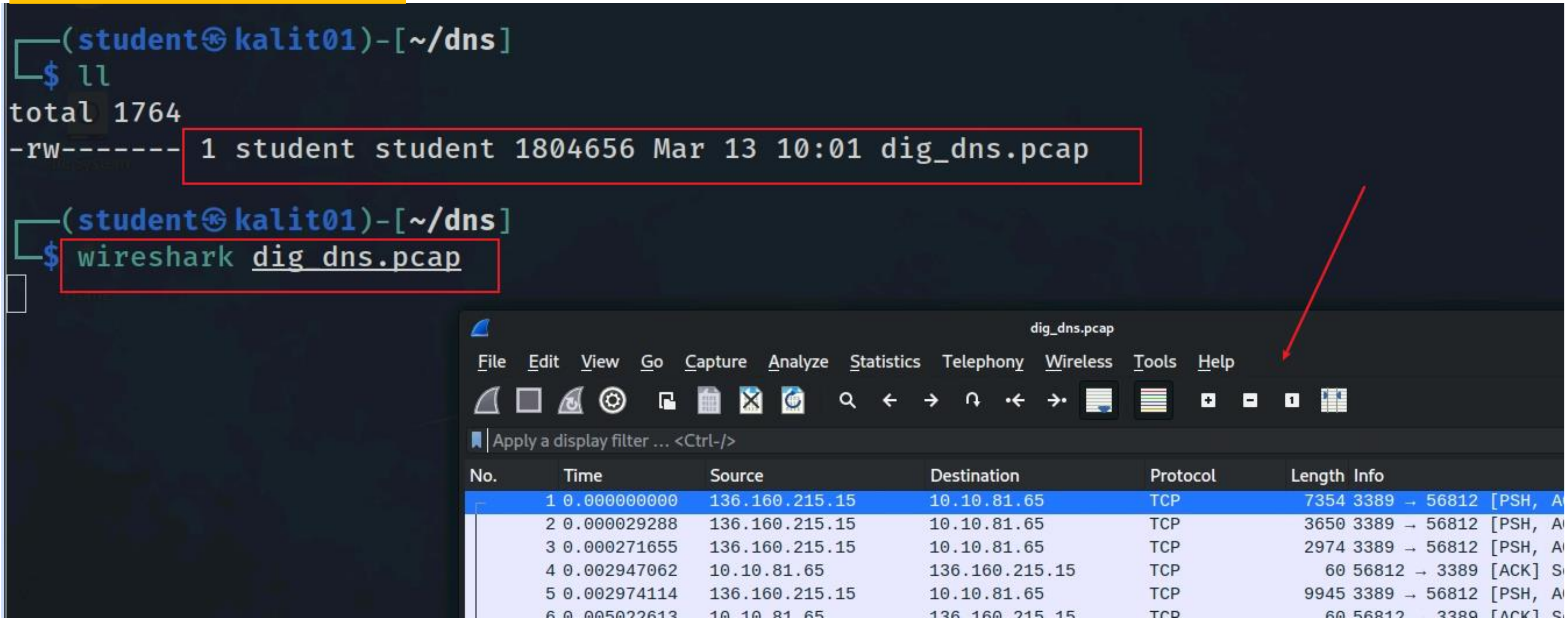
student@kali01: ~/dns 105x7

```
┌──(student㉿kalit01)-[~/dns]
└─$ tshark -i eth0 -w dig_dns.pcap
Capturing on 'eth0'
 ** (tshark:185262) 10:00:59.798575 [Main MESSAGE] -- Capture started.
 ** (tshark:185262) 10:00:59.798642 [Main MESSAGE] -- File: "dig_dns.pcap"
562 ^C
tshark:
```

1. Capturing traffic

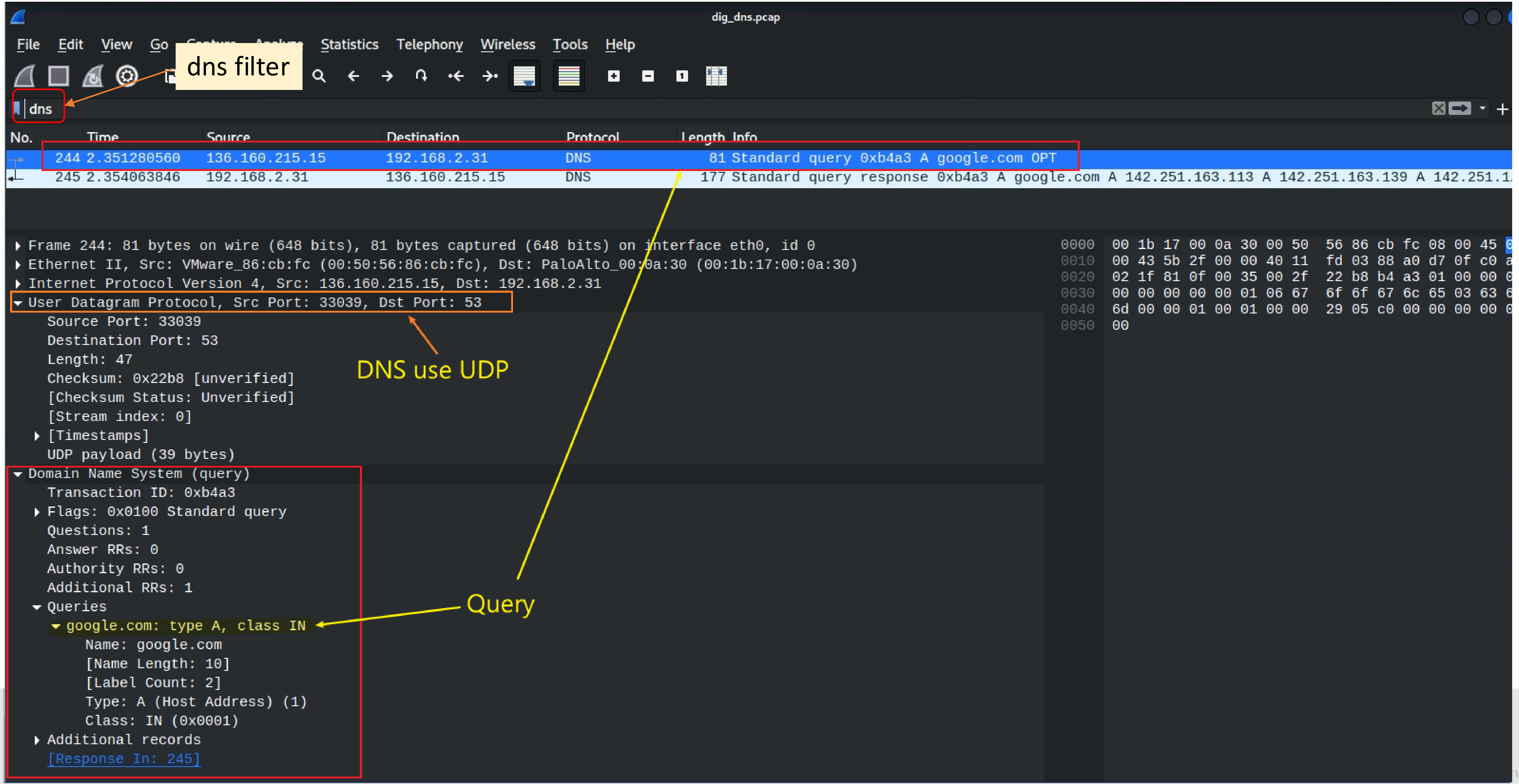
3. Control+c

View captured DNS traffic

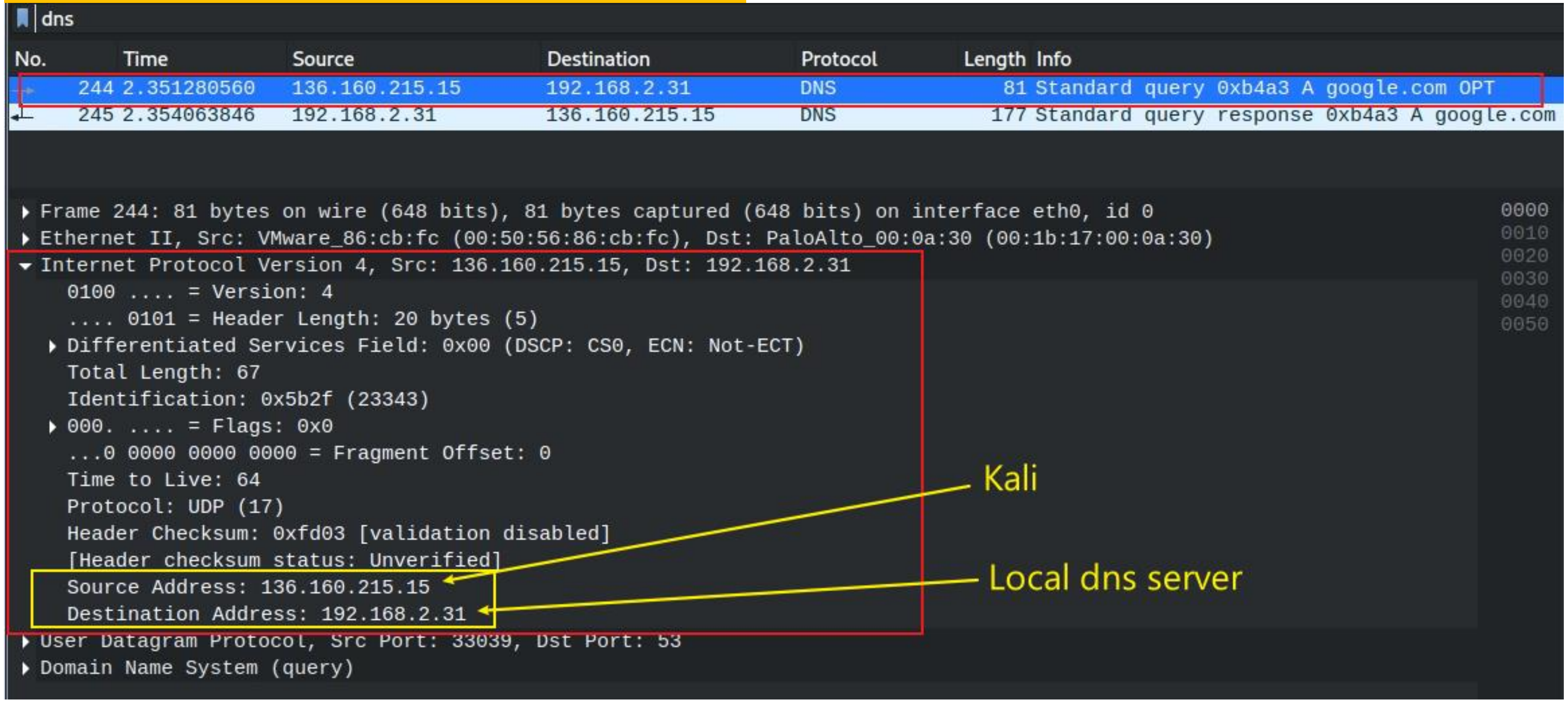


wget https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap

View layers DNS/UDP/IP/Frame of DNS query



View DNS query source and destination IPs



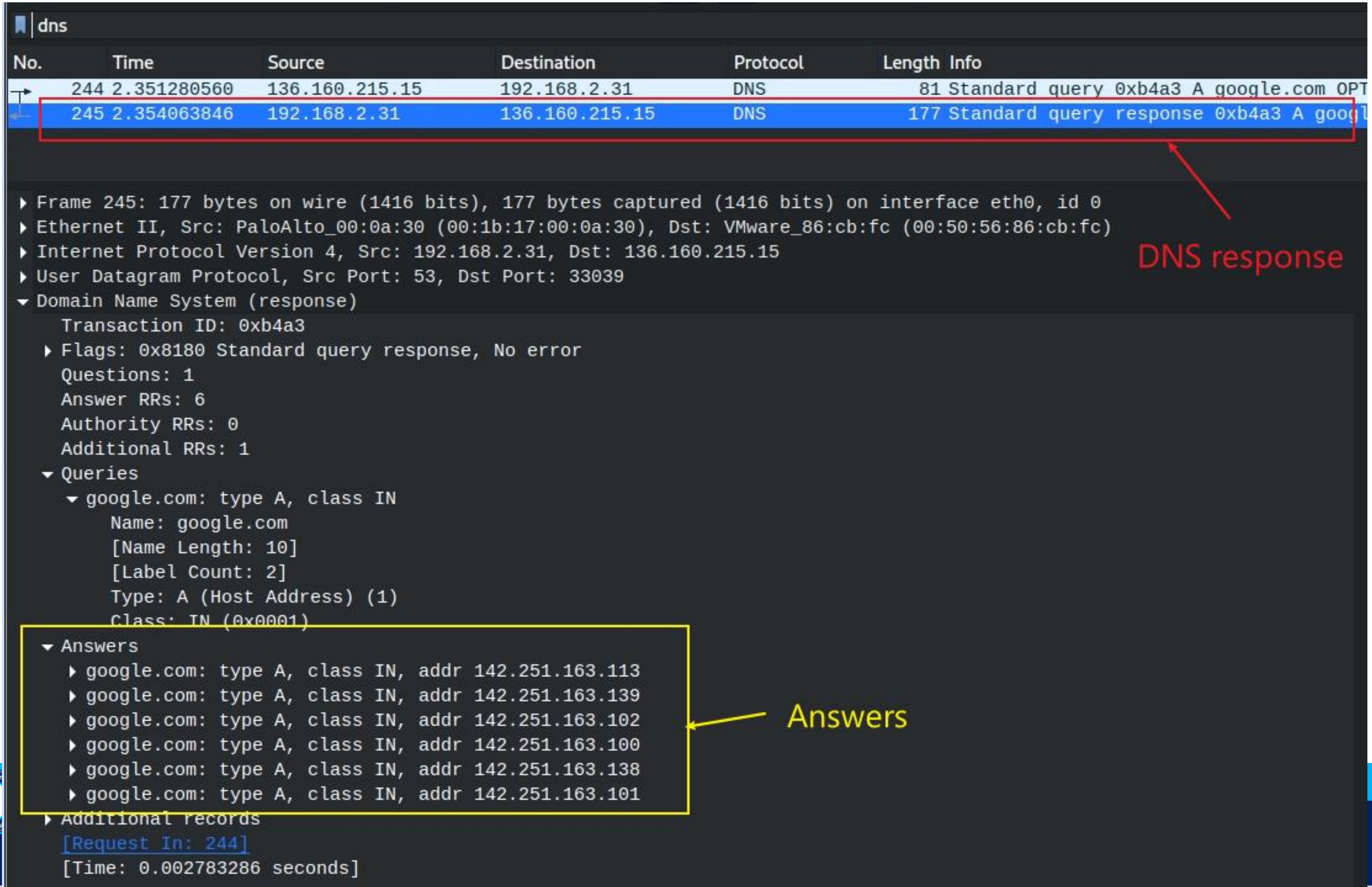
Local DNS configuration file

lists the DNS servers that your system is using for name resolution.

```
┌──(student㉿kalit01)-[~/dns]
└─$ cat /etc/resolv.conf
# Generated by NetworkManager
search lab.ubalt.edu
nameserver 192.168.2.31
nameserver 192.168.3.31
```

match DNS traffic

DNS response



One of answers

```
▾ Domain Name System (response)
    Transaction ID: 0xb4a3
  ▸ Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 6
    Authority RRs: 0
    Additional RRs: 1
  ▾ Queries
    ▾ google.com: type A, class IN
        Name: google.com
        [Name Length: 10]
        [Label Count: 2]
        Type: A (Host Address) (1)
        Class: IN (0x0001)
  ▾ Answers
    ▾ google.com: type A, class IN, addr 142.251.163.113
        Name: google.com
        Type: A (Host Address) (1)
        Class: IN (0x0001)
        Time to live: 25 (25 seconds)
        Data length: 4
        Address: 142.251.163.113
    ▸ google.com: type A, class IN, addr 142.251.163.139
    ▸ google.com: type A, class IN, addr 142.251.163.102
    ▸ google.com: type A, class IN, addr 142.251.163.100
    ▸ google.com: type A, class IN, addr 142.251.163.138
    ▸ google.com: type A, class IN, addr 142.251.163.101
  ▸ Additional records
    [Request In: 244]
    [Time: 0.002783286 seconds]
```

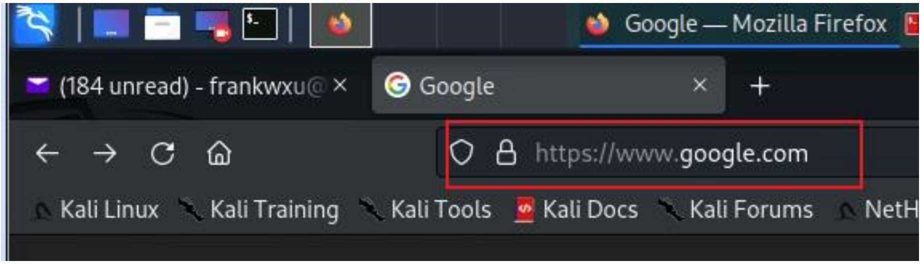
- maps the domain name to an IP address.
- refers to the Internet class.

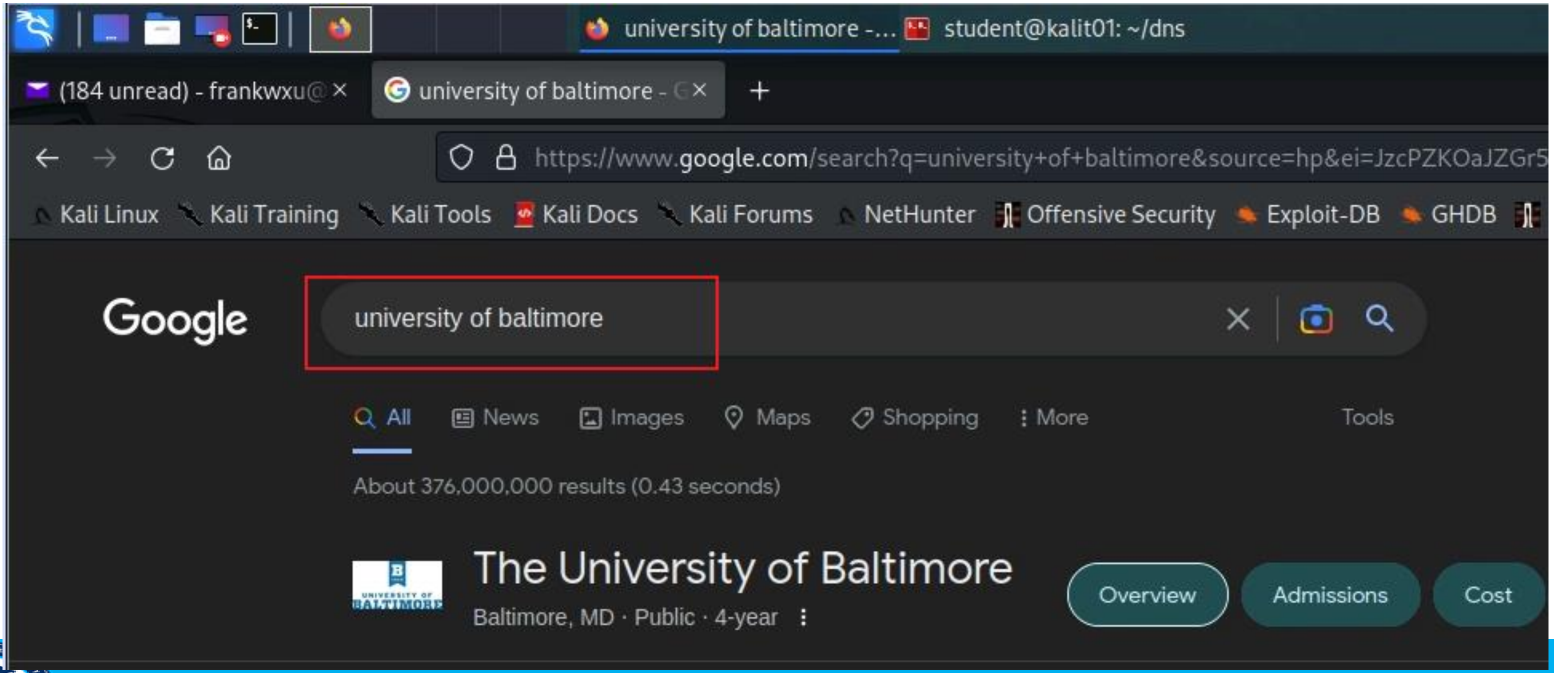
Observe the DNS traffic of accessing google.com

Command to capture traffic

```
student@kalit01: ~/dns 105x17
┌──(student㉿kalit01)-[~/dns]
└─$ tshark -i eth0 -w http_dns.pcap
Capturing on 'eth0'
 ** (tshark:186710) 10:45:52.135152 [Main MESSAGE] -- Capture started.
 ** (tshark:186710) 10:45:52.135310 [Main MESSAGE] -- File: "http_dns.pcap"
6897 ^C
tshark:
```

Accessing google.com and google search





Multiple pairs of DNS query and response because Google results page contains multiple hidden URLs (ads, images)

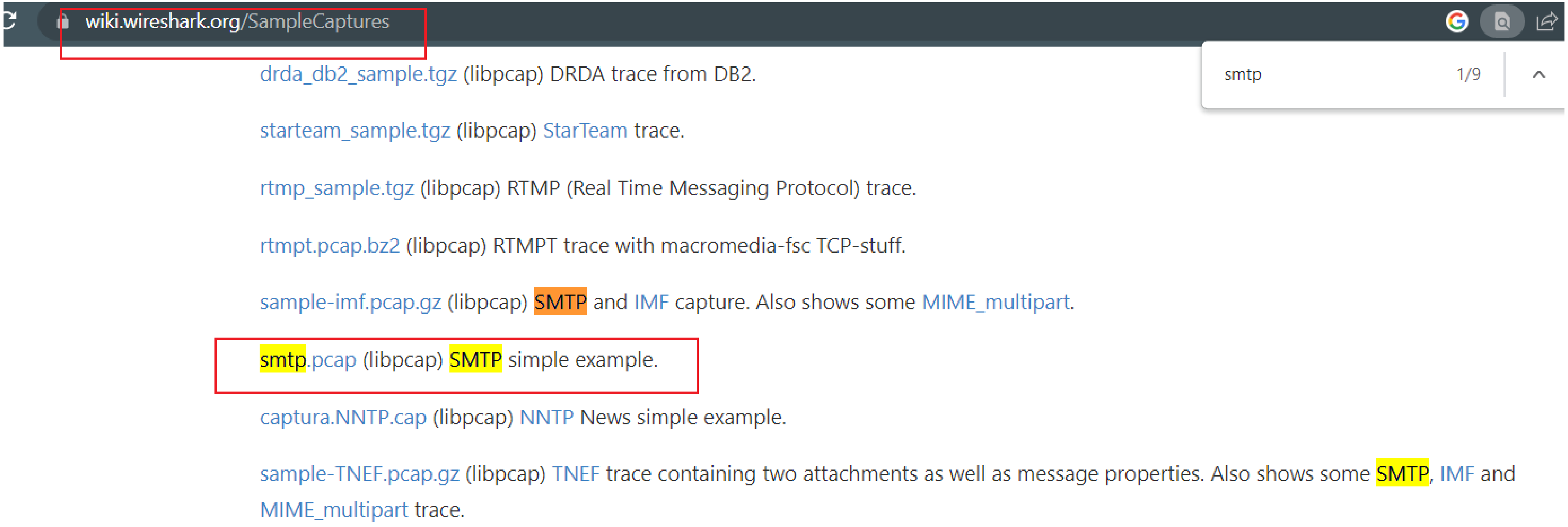
dns

No.	Time	Source	Destination	Protocol	Length	Info
448	7.001331416	136.160.215.15	192.168.2.31	DNS	85	Standard query 0x0b4c A www.google.com OPT
449	7.004296880	192.168.2.31	136.160.215.15	DNS	181	Standard query response 0x0b4c A www.google.com A 142.251.163.106 A 142.251.163.104 A 142.2
508	7.168967105	136.160.215.15	192.168.2.31	DNS	86	Standard query 0xcdc6 A www.gstatic.com OPT
509	7.171260202	192.168.2.31	136.160.215.15	DNS	102	Standard query response 0xcdc6 A www.gstatic.com A 172.253.63.94 OPT
877	7.597732573	136.160.215.15	192.168.2.31	DNS	86	Standard query 0x1fb2 A apis.google.com OPT
878	7.600481169	192.168.2.31	136.160.215.15	DNS	203	Standard query response 0x1fb2 A apis.google.com CNAME plus.l.google.com A 172.253.115.101
885	7.613979796	136.160.215.15	192.168.2.31	DNS	85	Standard query 0xb37e A ogs.google.com OPT
891	7.616809262	192.168.2.31	136.160.215.15	DNS	202	Standard query response 0xb37e A ogs.google.com CNAME www3.l.google.com A 142.251.163.139 A
1044	7.741376643	136.160.215.15	192.168.2.31	DNS	88	Standard query 0x788d A fonts.gstatic.com OPT
1046	7.743667806	192.168.2.31	136.160.215.15	DNS	104	Standard query response 0x788d A fonts.gstatic.com A 172.253.122.94 OPT
1292	8.032640338	136.160.215.15	192.168.2.31	DNS	91	Standard query 0xe579 A adservice.google.com OPT
1294	8.035004001	192.168.2.31	136.160.215.15	DNS	123	Standard query response 0xe579 A adservice.google.com A 172.253.63.154 A 172.253.63.155 OPT
1311	8.055027168	136.160.215.15	192.168.2.31	DNS	86	Standard query 0xff44 A play.google.com OPT
1312	8.057694788	192.168.2.31	136.160.215.15	DNS	182	Standard query response 0xff44 A play.google.com A 142.251.163.101 A 142.251.163.139 A 142.
1381	8.205834328	136.160.215.15	192.168.2.31	DNS	98	Standard query 0x10ab A googleads.g.doubleclick.net OPT
1383	8.208152910	192.168.2.31	136.160.215.15	DNS	130	Standard query response 0x10ab A googleads.g.doubleclick.net A 172.253.62.157 A 172.253.62.
1388	8.214932976	136.160.215.15	192.168.2.31	DNS	98	Standard query 0xa54d A safebrowsing.googleapis.com OPT
1389	8.217530617	192.168.2.31	136.160.215.15	DNS	114	Standard query response 0xa54d A safebrowsing.googleapis.com A 142.251.16.95 OPT
2025	17.716448151	136.160.215.15	192.168.2.31	DNS	97	Standard query 0xb5df A encrypted-tbn0.gstatic.com OPT
2026	17.719280345	192.168.2.31	136.160.215.15	DNS	193	Standard query response 0xb5df A encrypted-tbn0.gstatic.com A 172.253.115.113 A 172.253.115
2512	21.168919125	136.160.215.15	192.168.2.31	DNS	96	Standard query 0xc230 A lh5.googleusercontent.com OPT
2516	21.171259181	192.168.2.31	136.160.215.15	DNS	141	Standard query response 0xc230 A lh5.googleusercontent.com CNAME googlehosted.l.googleuserc
2542	21.194041420	136.160.215.15	192.168.2.31	DNS	84	Standard query 0x1ad7 A ocsp.pki.goog OPT
2543	21.196389535	192.168.2.31	136.160.215.15	DNS	135	Standard query response 0x1ad7 A ocsp.pki.goog CNAME pki-goog.l.google.com A 142.251.16.94
4248	31.819012899	136.160.215.15	192.168.2.31	DNS	84	Standard query 0x111f A id.google.com OPT
4249	31.821454919	192.168.2.31	136.160.215.15	DNS	100	Standard query response 0x111f A id.google.com A 172.217.30.195 OPT

Multiple pairs of query and response

DNS analysis of Email traffic

Lab Files



https://wiki.wireshark.org/SampleCaptures

Download smtp.pcap

```
┌──(kali㉿kali)-[~/smtp]
└─$ wget https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
--2023-03-10 14:58:08--  https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptur
es/smtp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org)... 172.67.75.39, 104.26.10.240, 104.26.11.240, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|172.67.75.39|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: ‘smtp.pcap’

smtp.pcap                100%[===================================>]  27.20K  --.-KB/s    in 0.004s

2023-03-10 14:58:08 (6.91 MB/s) - ‘smtp.pcap’ saved [27850/27850]
```

DNS query

Router, please find the IP of mail.patriots.in

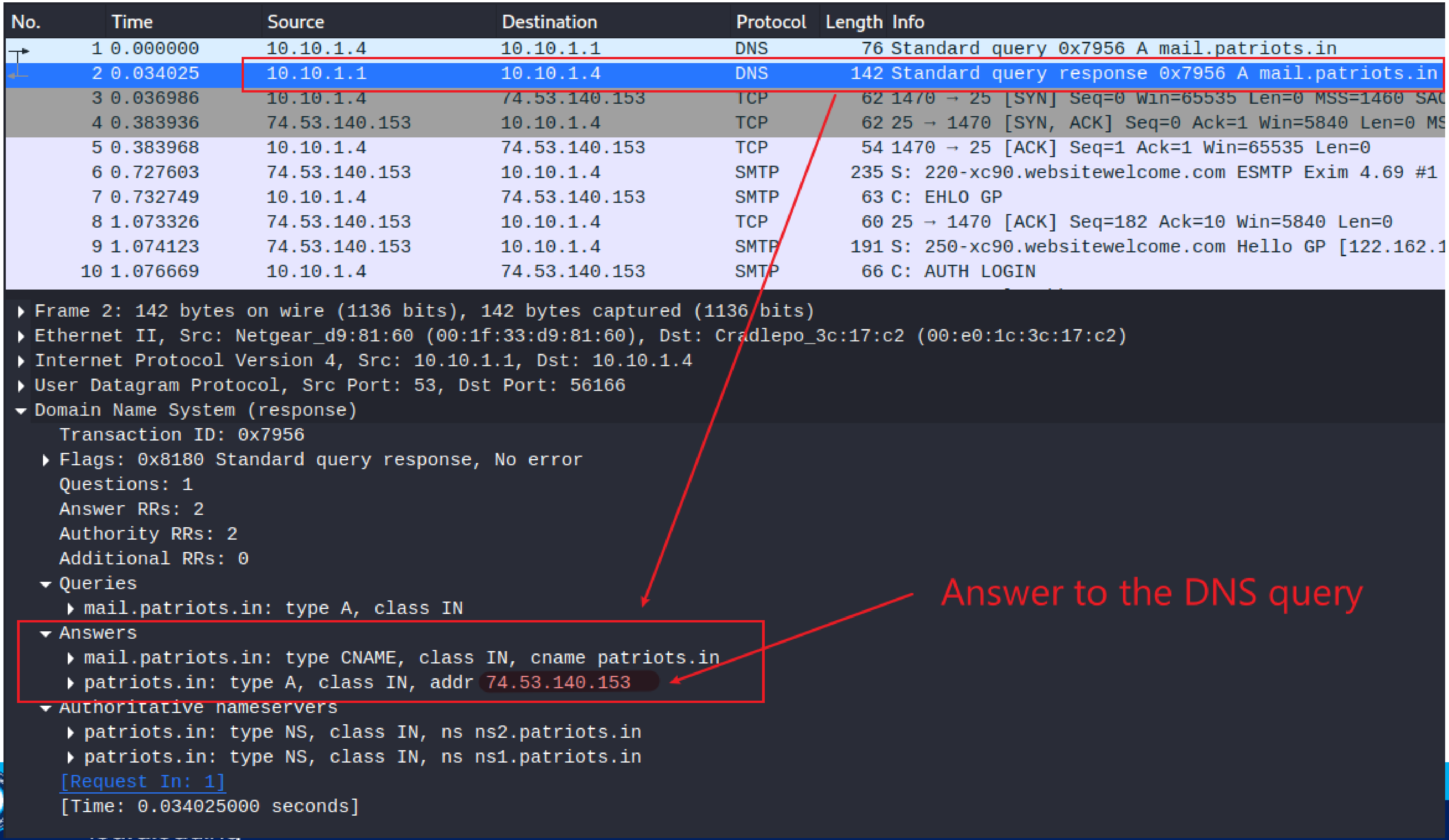
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.10.1.4	10.10.1.1	DNS	76	Standard query 0x7956 A mail.patriots.in
2	0.034025	10.10.1.1	10.10.1.4	DNS	142	Standard query response 0x7956 A mail.patriots.in
3	0.036986	10.10.1.4	74.53.140.153	TCP	62	1470 → 25 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SAC
4	0.383936	74.53.140.153	10.10.1.4	TCP	62	25 → 1470 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MS
5	0.383968	10.10.1.4	74.53.140.153	TCP	54	1470 → 25 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6	0.727603	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
8	1.073326	74.53.140.153	10.10.1.4	TCP	60	25 → 1470 [ACK] Seq=182 Ack=10 Win=5840 Len=0
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.1
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN

```
▸ Frame 1: 76 bytes on wire (608 bits), 76 bytes captured (608 bits)
▸ Ethernet II, Src: Cradlepo_3c:17:c2 (00:e0:1c:3c:17:c2), Dst: Netgear_d9:81:60 (00:1f:33:d9:81:60)
▸ Internet Protocol Version 4, Src: 10.10.1.4, Dst: 10.10.1.1
▸ User Datagram Protocol, Src Port: 56166, Dst Port: 53
▾ Domain Name System (query)
    Transaction ID: 0x7956
  ▸ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▾ Queries
    ▸ mail.patriots.in: type A, class IN
    [Response In: 2]
```

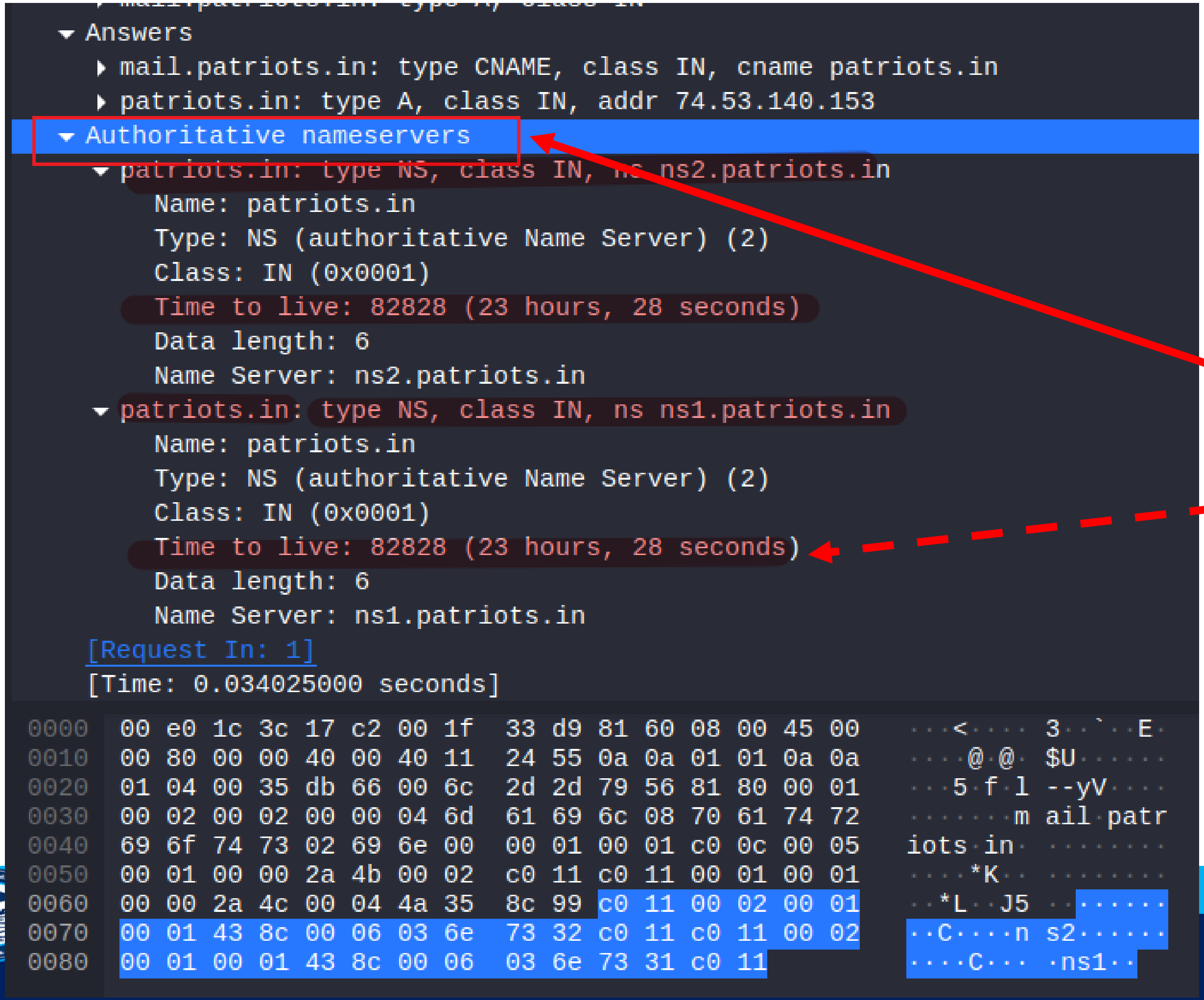
Application layer protocol

asking it to return the IPv4 (Type A) address associated with a given domain name, using the Internet namespace (Class IN).

DNS query response



Authoritative nameservers =authoritative DNS servers



- Answer from the router DNS caching
- The original and official information about a domain name from `ns1.patriots.in` or `ns1.patriots.in`.
- The DNS entry will expire
- Possible DNS attack

Handshaking (SYN, [SYN, ACK], ACK)